

Farshad Moradi Shahrababak

Cybersecurity Researcher

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Summary

MSc candidate in Computer Engineering with research interests in cybersecurity, privacy-preserving systems, DevSecOps, trustworthy AI and secure distributed software. Experienced in data anonymization, medical data protection and AI applications in healthcare, with additional industry expertise in software engineering, DevOps, and cloud-native platforms.

Education

MSc In Computer Engineering (Software Platforms and Cybersecurity curriculum) Sep 2023 to Expected: Sep 2026
University Of Genoa (Dibris Polytechnic Interschool Section)

- **Weighted course average:** 96/110 (thesis defense will add further points)
- **Thesis:** Designing a Defense-in-Depth Framework for Data De-Identification in Distributed Medical Imaging Systems
- **Selected coursework:** Computer Security, Data Protection and Privacy, Mobile Security, Binary Analysis, Machine Learning, Trustworthy AI

BSc In Computer Engineering (Software Engineering curriculum) Sep 2018 to Dec 2022
Azad University Tehran North Branch

- **GPA:** 15/20
- **Thesis:** COVID-19 Detection from Chest CT Scans Using CNNs

Research Experience

Designing a Defense-in-Depth Framework for Data De-Identification in Distributed Medical Imaging Systems

MSc thesis project, University of Genoa

Supervisor: Prof. Luca Verderame

Overview: Secure software engineering framework that integrates defense-in-depth, DevSecOps, and data de-identification to protect privacy and security in distributed medical imaging systems.

Contribution: Identify and assess vulnerabilities in distributed medical imaging systems, examine potential exploitation scenarios, and develop state-of-the-art security solutions to address these challenges.

Mondrian-Based Multidimensional K-Anonymity

Research project, University of Genoa

Supervisors: Prof. Gaspare Ferraro, Prof. Marina Ribaudó.

Overview: Implemented a data anonymization algorithm using the Mondrian approach to achieve k-anonymity in multidimensional datasets.

Contribution: Developed Python implementation of the Mondrian algorithm; ensured flexible handling of multiple quasi-identifiers and tested anonymization on sample datasets.

COVID-19 Detection from Chest CT Scans Using CNNs

BSc thesis, Islamic Azad University Tehran North

Supervisor: Dr. Arzhang Malek

Overview: Designed and trained convolutional neural network models for binary classification of COVID-positive versus healthy chest CT scans.

Contribution: Designed and trained a custom CNN architecture from scratch in Python (TensorFlow/Keras) for binary classification of COVID-positive versus healthy chest CT scans; achieved approximately 93% test accuracy.

Publications

In preparation:

Designing a Defense-in-Depth Framework for Data De-Identification in Distributed Medical Imaging Systems

Supervisor: Prof. Luca Verderame

Target venue: 31st Nordic Conference on Secure IT Systems (NordSec 2026), Aarhus, Denmark

Expected submission: Aug 17, 2026

Teaching Experience

Teaching Assistant, Advanced Programming

Sharif University of Technology, Faculty of Industrial Engineering, Tehran, Iran

- Assisted Prof. Moslem Habibi and Prof. Majed Nabavian in delivering a programming course for industrial engineering students, covering object-oriented Python, JavaScript, Django, Git, Docker, and relational databases.
- Mentored students transitioning into software roles at startups, focusing on industry-grade development practices (version control, containerization, testing).

Professional Experience

Software Engineer

Feb 2024 to Mar 2025

Ericsson (Genova, Italy)

- Automated **Builds**, **Smoke tests**, and **Deployment** pipelines reduced deployment time by 50%.
- Applied **blue-green deployment practices** to achieve **99% uptime** and **reduce downtime during releases**.
- Established **continuous monitoring** and **automated rollback** triggers using **observability tools**.

Software Engineer

Feb 2021 to Nov 2023

FANAP (Tehran, Iran)

- **Launched an ERP PWA** using **DevOps practices** for **deployment** and **monitoring**.
- Worked closely with managers to **gather requirements** and **translate business needs into ERP solutions**.
- Led weekly **code reviews**, **mentored 5+ junior developers**, and **reduced code duplication** by 20%.

Skills

Programming Languages: Python, Java, JavaScript, C/C++

Cybersecurity

- **DevSecOps:**
 - **Threat Modeling:** STRIDE, DREAD, MITRE ATT&CK; applied to CI/CD pipeline threats (Poisoned Pipeline Execution, secret exposure, dependency confusion)
 - **Application Security Testing:** SAST (SonarQube, Snyk Code), DAST (OWASP ZAP, Burp Suite), SCA (Snyk, OWASP Dependency-Check), IAST concepts
 - **Secrets & Identity:** HashiCorp Vault, GitHub Secrets, least-privilege scopes
- **Trustworthy & Adversarial Machine Learning**
 - **Evasion attacks:** Carlini-Wagner, DeepFool, PGD, transfer-based attacks
 - **Poisoning & backdoor attacks:** BadNets, neural trojans, clean-label attacks
 - **Availability attacks:** sponge examples
 - **Privacy attacks on ML:** membership inference, model extraction
 - **LLM-specific threats:** prompt injection (direct/indirect), jailbreaking, training data extraction
 - **Defenses:** adversarial training, certified robustness, input rejection, differential privacy (DP-SGD)
 - **Frameworks:** SecML, ART, Foolbox
- **Cryptography**
 - **Symmetric:** AES-GCM/CBC, ChaCha20-Poly1305, HMAC
 - **Asymmetric:** RSA, ECC (Curve25519, P-256), Diffie-Hellman, ECDH
 - **Hashing & KDFs:** SHA-2, SHA-3, BLAKE2, Argon2, HKDF
- **Binary Analysis:** Ghidra, Debugging, Reversing, BOF and Shellcoding, Exploitation
- **Data Protection and Privacy:** Anonymization, Pseudonymization, HIPAA, NIST, GDPR

AI: Machine Learning, Deep learning frameworks (PyTorch, TensorFlow), LLM frameworks (LangChain, LlamaIndex), RAG

DevOps: Linux, Bash, Git, CI/CD (Jenkins, GitHub Actions, GitLab CI, ArgoCD), containers and orchestration (Docker, Kubernetes, Helm), IaC (Terraform, Ansible), monitoring (Prometheus, Loki, Grafana), AWS

Languages

Persian (native), English (C1), Italian (A1)